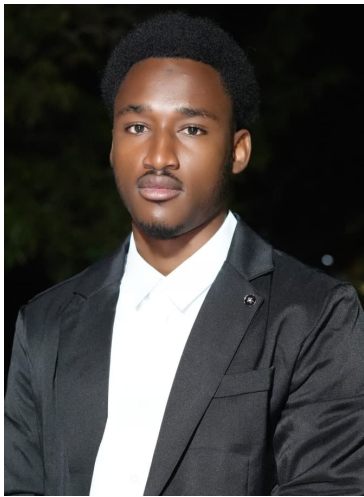




SEYDOU DIALLO

Software Developer | AI/NLP Enthusiast | Linux Administrator |
Community Builder

- **Address:** Somone, Senegal
- **Email:** mail.seydou.diallo@gmail.com
- **GitHub:** [sudoping01](#)
- **HuggingFace:** [sudoping01](#)
- **LinkedIn:** [seydou diallo](#)

PROFILE

Software developer and AI/NLP researcher building speech and language technology for African languages - ASR, TTS, machine translation, and LLMs. Founder of MALIBA-AI and co-founder of DJELIA, where I take models from data collection through production deployment. First-author work on Bambara language technology published at peer-reviewed venues, with open-source models released on Hugging Face and developer SDKs in production.

EDUCATION

Dakar American University of Science and Technology | Dakar, Senegal
Computer Science | 2024 - Present

Ecole Normal D'Enseignement Technique et Professionnel | Bamako, Mali
Electronic Engineering | 2023 - 2024

Technical Baccalaureate in Electronic Engineering | Bamako, Mali
Industrial Sciences and Technologies | July - 2022

First National in Malian Technical Baccalaureate session

Specialization: Electronic Engineering

PUBLICATIONS

BambaraMLLM: A Unified Multilingual Multimodal Large Language Model for Comprehensive Bambara Language Processing

Seydou Diallo, Allahsera Auguste Tapo, Kevin Assogba, Christopher Homan - **First author**

- Unified multimodal model covering ASR, machine translation, text generation, and TTS for Bambara in a single framework

- Built a linguistically informed instruction-data framework injecting Bambara morphology, syntax, and cultural context to generate 2M+ conversational instruction examples
- [Paper](#)

Where Are We At with Automatic Speech Recognition for the Bambara Language?

Seydou Diallo, Yacouba Diarra, Mamadou K. Keita, Panga Azazia Kamaté, Adam Bouno Kampo, Aboubacar Ouattara - First author

- Introduced the first standardized ASR benchmark for Bambara, built on professionally recorded Malian constitutional text with manual segmentation, alignment, and quality review
- Evaluated 37 systems and released a public leaderboard for transparent Bambara speech-technology evaluation
- arXiv: [2602.09785](#)

InstructLR: A Scalable Approach to Create Instruction Dataset for Under-Resourced Languages

Mamadou K. Keita, Sebastien Diarra, Christopher Homan, Seydou Diallo - Co-author

- Framework combining LLM generation, RAG-based correction, and human-in-the-loop validation to build high-quality instruction datasets for low-resource languages
- Produced the 50k-scale BambaraInstruct, ZarmaInstruct, and FulfuldeInstruct benchmarks
- arXiv: [2512.02213](#)

GAIFE: Generative AI for Education (NAACL 2025 Findings) - Co-author

- Human-in-the-loop workflow using LLMs, machine translation, and linguistic tools to produce culturally grounded children's reading materials in Bambara, a low-resource language
- Built a learning library of 174 illustrated books (~4,000 pages, 850+ original images), multiplying Bambara children's content online roughly tenfold
- Pilot reading programs achieved a 67% reduction in children unable to read Bambara
- [Paper](#) | [Library](#)

PROFESSIONAL EXPERIENCE

PROF

Co-Founder | [DJELIA](#)

2024 – Present

- Own the full AI model lifecycle for Bambara - data collection, training, evaluation, and deployment
- Built and maintain production-grade ASR, TTS, MT, and LLM models for Bambara, served via cloud APIs
- Shipped Python and JavaScript SDKs so third-party applications can integrate Djelia's language models
- Turned research-stage language models into reliable, documented services for local developers

Founder | [MALIBA-AI Community](#)

2025 – Present

- Founded and lead a community-driven AI initiative for indigenous Malian languages, owning research from data collection to final model
- Built and maintain ASR, TTS, LLM, and MT models across multiple Malian languages
- Released inference SDKs and serving infrastructure so developers and communities can integrate MALIBA-AI models directly
- Designed speech-first interfaces for primarily oral languages, expanding AI access in education, healthcare, agriculture, and local business

Software Developer & AI Engineer | [Caytu Robotics](#)

2024 – Present

- Built and maintain a scalable IoT ecosystem for remote control, real-time video streaming, and data monitoring
- Developed Caytu Avatar, a multimodal AI agent enabling natural-language interaction with any REST API, including the Caytu IoT ecosystem
- Created tools and SDKs for integrating AI services, automation workflows, and assistant features

Robotics Team Member | Malian National Team

2021, 2023

- Represented Mali in international robotics competitions
- Collaborated on complex engineering projects and problem-solving challenges

RESEARCH PROJECTS

Large Language Models

MALIBA-LLM | Lead Developer | MALIBA-AI

- Built **MALIBA-LLM**, the first open-source large language model for Bambara (Bamanankan), fine-tuned from google/gemma-3n-E2B-it
- Supports Bambara ↔ French/English code-switching, instruction following, translation, and Mali-specific knowledge
- Reduced validation loss to **0.4952** (93.4% lower than the 7.4595 starting point)
- Released on Hugging Face for education, translation, and conversational use: [maliba-llm](#)

Speech Synthesis

Bambara Text-to-Speech | Lead Developer | MALIBA-AI

- Open-source Bambara TTS system rated **4.2/5.0** for quality and **4.1/5.0** for naturalness by 10 native Bambara speakers
- Targets the digital exclusion affecting millions of non-literate adults in sub-Saharan Africa
- Released on Hugging Face - **8,712 downloads** as of October 2025: [MALIBA-AI/bambara-tts](#)

MALIBA-TTS: Multilingual Speech Synthesis for Malian Languages | *Lead Contributor*

- Developed **text-to-speech (TTS) models** for six Malian languages: **Bambara, Boomu, Dogon, Pular, Songhoy, and Tamasheq**
- Based on **Meta's MMS-TTS** and **VITS**, optimized for **CPU inference** and **real-time synthesis**
- Enables **voice interfaces, education, and accessibility tools** across Mali
- Promotes **language inclusion and digital accessibility** under the *MALIBA-AI* initiative
- Models: huggingface.co/MALIBA-AI/malian-tts

GAIFE: Generative AI for Education

Co-Author | Bill & Melinda Gates Foundation Grand Challenges | RobotsMali

- Co-developed a novel methodology combining generative AI and automatic translation to produce educational content at scale for Bambara, a low-resource language
- Co-author on the resulting peer-reviewed paper (NAACL 2025 Findings)
- Applied the methodology to build a library of 174 published illustrated books, contributing Bambara linguistic review for accuracy, fluency, and cultural relevance
- [Paper](#) | [Library](#)

Bayelemabaga: Bambara Machine Translation

Data Processing | RobotsMali | RIT Partnership

- Contributed to the first machine translation model for the Bambara–French language pair
- Handled extraction, cleaning, and alignment of the parallel corpora behind the model
- [Project Details](#)
- [Dataset](#)

Automatic Speech Recognition Systems

Wolof ASR Models | Lead Developer | CAYTU ROBOTICS

- Built **whosper-large-v2** and **whosper-large**, open-source Wolof ASR models
- Handle multilingual input across Wolof, French, and English
- whosper-large-v2 reached **24% WER** and **11% CER** on a 9-hour evaluation set
- **3,893 downloads** (whosper-large-v2) and **679 downloads** (whosper-large) in first week
- Models: [whosper-large-v2](#) | [whosper-large](#)

Bambara ASR Models | Lead Developer | MALIBA-AI

- Built the **maliba-asr** series for Bambara ASR
- maliba-asr-v2 reached **24.33% WER** and **12.45% CER** on a 3-hour evaluation set
- Handles Bambara–English code-switching
- Latest Model : [bambara-asr-v1](#)

Songhay ASR Model | | Lead Developer | MALIBA-AI

- Built **songhay-asr-v1**, an ASR model for the Songhay language
- Reached **16.58% WER** and **4.63% CER** on the test set
- Supports bilingual Songhay–French transcription

- Model: [songhoy-asr-v1](#)

IoT Assistant in Local Languages

Lead Developer | Caytu Robotics

- Built an IoT assistant operating fully in Wolof
 - Designed a scalable IoT ecosystem with remote control and real-time streaming
 - Added workflow automation and multilingual support
 - Platform: [caytu.link](#)
 - Open-sourced architecture at Ndabax Senegal 2024
-

OPEN SOURCE CONTRIBUTIONS

Software Development Kits (SDKs)

NabooPay Payment Gateway SDKs | *Lead Developer & Maintainer*

- Python SDK for Senegalese mobile money payments (Wave, Orange Money, Free Money)
- Node.js SDK in TypeScript with full type definitions and automatic retry logic
- Transaction management and cashout operations, with both sync and async support
- Code Source: [Python SDK](#) | [Node.js SDK](#)

Djelia AI Language Platform SDKs | *Lead Developer & Maintainer*

- Python SDK: Complete SDK for Bambara language AI (translation, ASR, TTS) with streaming support
- JavaScript SDK: Modern implementation for web and Node.js environments
- Features: Multi-language translation, real-time transcription streaming, natural TTS with speaker descriptions
- Code Source: [Python SDK](#) | [JavaScript SDK](#)

MALIBA-AI Bambara TTS SDK | *Lead Developer & Maintainer*

- Inference SDK for Bambara text-to-speech synthesis
- Production-ready inference system with 10 authentic speaker voices
- Optimized for educational platforms, accessibility tools, and voice interfaces
- Comprehensive documentation with usage examples and best practices
- Code Source: [GitHub Repository](#)

Infrastructure & Deployment Tools

Realtime Video/Audio WebRTC Streaming

- Enterprise-grade streaming solution containerized for easy deployment
- Enables low-latency cloud transmission from various sources
- Applications in remote monitoring, surveillance, telehealth, industrial IoT
- Code source: [GitHub](#) | [Docker Hub](#)

Wolof TTS Inference Server

- Deployment server for a Wolof speech synthesis system
- Makes voice technology available for Wolof, a primary language of Senegal
- Code Source: [GitHub Repository](#)

Adia_TTS Segmentation Algorithm

- Turned a research model into a production-ready system
- Removed character-length limits while preserving voice quality
- Merged into the official model documentation
- Code Source: [GitHub](#) | [HuggingFace Discussion](#)

Additional projects: [GitHub Profile](#)

PREVIOUS VENTURES

KAALISI | Founder

AI-powered investment evaluation platform

- Developed scoring models assessing startups on financial performance, environmental impact, pandemic resilience, and youth employment
- Aimed to democratize access to capital through intelligent matchmaking

JAYAS Drone | Co-Founder

Drone-as-a-Service platform

- Built ready-to-fly drones and service infrastructure
-

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C/C++, Bash, Java

AI/ML Frameworks: PyTorch, Hugging Face Transformers, Whisper, FastAPI, Unsloth, Axolotl

Technologies: Docker, WebRTC, IoT, AWS, GCP, Linux Administration, git, Github

NLP Specializations: Speech Recognition (ASR), Text-to-Speech (TTS), Machine Translation (MT), Large Language Model (LLM)

Languages: Bambara (Native), French (Fluent), English (Fluent)

COMMUNITY LEADERSHIP

MALIBA-AI - Founder & Lead

Building an AI ecosystem for indigenous Malian languages, centered on speech-first technology and

accessibility, with applications across education, healthcare, and business.

[Introduction to MALIBA-AI](#) | [Hugging Face Community](#) | [Github](#)

References

Sebastien DIARRA - ML/AI Researcher

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